2025 International Conference on Advancement in Data Science, E-learning and Information System (ICADEIS) | 979-8-3315-1332-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICADEIS65852.2025.10933065

Carfin: Car Recommender System Based on Conversational Recommender System Using Large Language Model

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***Abstract*—Various approaches have been used to develop car recommender systems, but limitations remain in personalizing information. Existing recommender systems often provide less relevant and flexible recommendations because they do not fully understand users' needs. Therefore, there is a need for a car recommender system that can offer more personalized recommendations, understand user needs better, and allow for more flexible interactions between the system and the user. This research aims to address these limitations and open up opportunities to enhance the performance and personalization of recommender systems in the automotive field. We developed a car recommender system named Carfin, which utilizes a Large Language Model (LLM) based Conversational Recommender System (CRS) to provide more accurate recommendations that align with user preferences and allow more flexible interactions between the system and the user in the process of getting car recommendations. We performed fine-tuning on the GPT-4 model and prompted engineering to generate accurate recommendations. After fine-tuning, the model was implemented to interact with users through the Telegram platform. We then evaluated the model using BERT Precision, BERT Recall, BERT F1-Score, and Cosine Similarity metrics. The evaluation results showed a significant improvement in performance before and after fine-tuning. The evaluation of each BERT Precision metric increased from 0.8048 to 0.8541, BERT Recall from 0.8639 to 0.9183, BERT F1-Score from 0.8332 to 0.8850, and Cosine Similarity from 0.6590 to 0.8372. Based on these performance improvements, Carfin can provide personalized and relevant car recommendations through conversational interactions while focusing on the automotive domain.**

***Index Terms*—Car Recommender System, Large Language Model, Conversational Recommender System, Preferences Elicitation**

I. Introduction

The increase in car sales in Indonesia is driven by the growing demand for mobility and the public's preference for private vehicles. One of the challenges the automotive industry faces is consumer confusion due to the many car options available. In today's era, where technology is becoming increasingly widespread across various aspects of daily life, essential roles such as recommender systems provide information tailored to individual preferences [1]. As technology advances, the application of deep learning in recommender systems can enhance personalization, enabling the system to process more complex data and provide more relevant recommendations to user preferences [2].

Various studies have applied the Large Language Model (LLM) across different fields, including recommender systems. Huang et al. developed an LLM for automatic and cross-modal dental diagnosis in dentistry [3]. Taylor et al. introduced Galactica, an LLM for scientific knowledge capable of integrating and reasoning with scientific data [4]. Additionally, research on Conversational Recommender Systems (CRS) also shown significant progress, such as the work of Ayundhita et al., who developed a CRS for laptop recommenders based on functional needs [5].

Research on car recommender systems has also been growing. Boteju et al. developed a car recommender system using a Hybrid Recommender Algorithm method, combining Collaborative Filtering and Natural Language Processing to generate more personalized recommendations based on user preferences [1]. Nuanmeesri et al. implemented the SMOTE method and Random Forest to improve the accuracy of second-hand car recommender systems [6]. Meanwhile, Syril et al. used User-Based Collaborative Filtering for a car recommender system based on the user's geographical location [7]. Additionally, Suryadi et al. [21] conducted research on a car recommender system with an Ontology-based approach, showing an overall system performance evaluation and user satisfaction rate of 87.84%, as well as receiving positive feedback from users regarding their satisfaction with the recommender system.

Although various approaches have been used to develop car recommender systems, there are still areas for improvement such as limitations in personalizing information and user interaction. Therefore, this research proposes a recommender system named Carfin, which utilizes a Large Language Model (LLM) in a Conversational Recommender System (CRS). This approach allows the system to provide more relevant recommendations, better understand user needs, and offer more flexibility in the interaction between the system and the user [8] [9]. Carfin is designed to enhance the user experience through better natural language understanding and deeper personalization, helping users make more informed decisions.

This study aims to address the limitations of the Carfin car recommendation system, designed to enhance personalization and user interaction in automotive recommenda-

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tions. By Integrating Large Language Model (LLM) and Conversational Recommender System (CRS), we aim to improve recommendation accuracy and create more natural interactions between users and the system. However, this study has several limitations, including its exclusive focus on car recommendations, thus not covering recommendations for other automotive products. Additionally, the testing in this study is solely conducted to evaluate the performance of the LLM approach in CRS. Nevertheless, this research is expected to improve the performance and personalization capabilities of recommender systems in the automotive domain.

This paper discusses five main interconnected topics in the development of a car recommender system based on CRS using LLM It begins with Section I Introduction, which explains the background of the increase in car sales in Indonesia and the need for a recommender system to address consumer confusion. Section II Related Work, reviews various studies related to the implementation of LLM and CRS in recommender systems. Section III Methodology, provides detailed information on the system development, including LLM CRS implementation on recommender system, flow of system interaction, preprocessing, fine-tuning, and prompt engineering. Section IV Results and Discussion, explains the significant improvements in evaluation metrics after the fine-tuning process. The paper concludes with Section V Conclusion, which summarizes the successful development of the car recommender system and suggestions for future development.

II. Related Work

Several studies are related to using Large Language Model (LLM) and Conversational Recommender Systems (CRS) to develop recommender systems. The Safety Fine Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) approaches have been used to control LLM and improve their usability and accessibility across various domains [10]. An example of LLM application in healthcare is GatorTronGPT, which was trained using 277 billion words from clinical and general texts. GatorTronGPT has been shown to enhance biomedical natural language processing and generate high-quality synthetic text [11]. Similar studies also demonstrate that LLM can be applied across various domains to achieve better results through personalization.

A new approach in CRS using SASRec and BERT models has been applied to predict topics, recommend items, and generate responses based on user preferences [12]. Conversational strategies such as rapport-building and persuasive explanations have been shown to improve users' perceptions of CRS [9]. Additionally, Permana et al. conducted research showing that when users interact with a CRS implementing rapport-building, there is a 25% higher satisfaction rate and a 20% increase in conversion rates [20]. This strategy demonstrates the potential of CRS to provide more intuitive and user-centric recommendations.

LLM models such as Flan-T5-Large and LLaMA-7b have shown advantages in improving CRS performance by managing sub-tasks, collaborating with expert models, and generating more relevant responses [13]. The collaboration between LLM and CRS has demonstrated significant improvements in pre-sales dialogues in e-commerce, with more informative responses [8]. Other studies indicate that user-based models can understand conversation preferences to generate more relevant recommendations [14].

The Hybrid Recommender Algorithm approach, which combines Collaborative Filtering, Knowledge-based Filtering, and Natural Language Processing (NLP), has resulted in a vehicle recommender system with an accuracy of up to 96% [1]. SMOTE-based and Random Forest-based systems have also been used to recommend used cars, achieving an accuracy of 98.84% by considering car specifications and consumer profiles [6]. Luyen Le N et al. developed a system that helps users find vehicles according to their profiles and preferences, simplifying the information search process [15].

III. Methodology

In this research, we developed a car recommender system named Carfin which utilizes Large Language Model (LLM) technology and Conversational Recommender System (CRS). Carfin is designed to help users find cars that match their needs and preferences through natural conversational interactions. We use an LLM model from OpenAI, which is expected to be effective in various conversational tasks [8] [10] [16]. In addition to LLM, Carfin also implements CRS to enable more natural interactions between the system and the user. Through the CRS approach, Carfin is expected to gradually understand user preferences through conversation and provide increasingly personalized car recommendations [12] [17]. The system will utilize a dataset that will be processed to train the model in generating appropriate responses and improving recommendation performance.

By combining LLM and CRS, Carfin is expected to collect, analyze, and process user data to provide structured and relevant car recommendations [6] [15]. This process will involve gathering information about user needs and preferences through conversation, data modeling, and providing suitable car recommendations. Through this approach, Carfin is expected to become an effective and user-friendly solution for users looking for the right car that matches their desires.

Our methodology addresses three key limitations of existing car recommender systems through specific technical approaches. First, to overcome limited automotive domain understanding, we perform fine-tuning on the GPT-4 model using 200 car-specific dialogue examples, enabling more accurate car-related responses. Second, to enhance personalization, we implement prompt engineering that guides the model to consider specific car attributes and user preferences when generating recommendations. Finally, to improve user interaction, we develop a conversational interface on Telegram that enables natural dialogue flow and dynamic responses. These approaches work together to create a more effective and user-friendly car recommendation system.

A. Flow of System Interaction

The interaction flow of the car recommender system begins when the user inputs preferences related to the desired car. Figure 1 shows the available car dataset undergoing a preprocessing stage to prepare the relevant data. Next, the GPT-4o model processes the user input by performing

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tokenization, embedding, and contextual analysis, combining the user's preferences with the dataset. The system generates car recommendations based on this analysis. If the initial recommendations are not suitable, the user can update their preferences, and the system will reprocess until the recommendations match the user's needs.

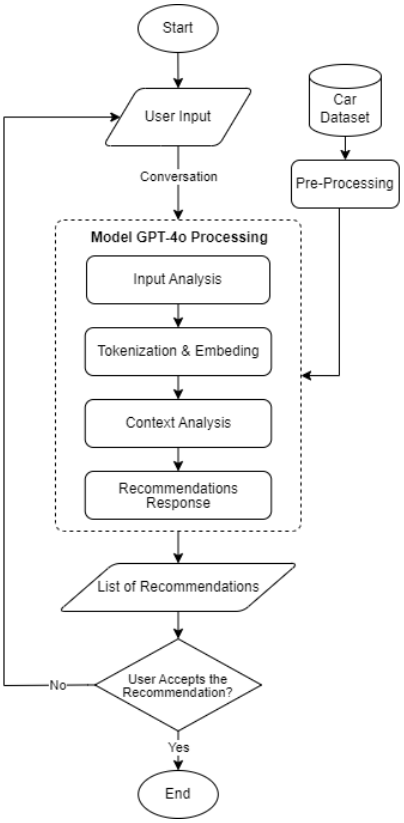


Fig. 1. System interactions

B. Preprocessing

This study uses a dataset from Kaggle containing car sales information from the OLX platform, covering 9,000 car types and models. In the preprocessing stage, irrelevant attributes, missing (null) values, and duplicate data are removed to ensure clean and accurate data. Table I shows the selected attributes focus on key features that support relevant car recommendations. Additionally, the data format is checked for consistency to ensure it can be used in model fine-tuning. This process ensures that the dataset is ready to be used by the model to generate recommendations that align with user preferences [18].

C. Fine-Tuning

The fine-tuning process begins with the creation of a prompt message for the user and the system in JSON format. Table II shows an example of the prompt for fine-tuning. In this stage, we will use 200 data divided into two parts where 160 data for training and 40 data for validation. This split is done to ensure that the model can learn effectively from the training data while also testing its performance on previously unseen validation data. This is important to prevent overfitting and ensure that the model can generalize well to new data. Fine-tuning is carried out using OpenAI, resulting in a domain-specific model for cars that maximizes the system's ability to provide accurate and relevant responses regarding car information [19].

TABLE I
REQUIRED RECOMMENDATION DATA

Attribute	Description
Car Name	A comprehensive identifier encompassing the manufacturer's brand, specific model designation, and variant details, providing a complete overview of the vehicle's identity.
Year	The manufacturing year of the vehicle, indicating when it was produced and helping assess its age and generation in the model lineup.
KMs driven	A crucial metric indicating the vehicle's total traveled distance, serving as a key indicator of usage intensity, maintenance requirements, and overall mechanical condition.
Fuel	The vehicle's required energy source specification, which could be gasoline, diesel, hybrid, electric, or alternative fuel types, affecting operating costs and environmental impact.
Transmission	The type of transmission of the car, determining the gearshift transmission of the car. It can be automatic or manual.
Car Features	A comprehensive inventory of the vehicle's built-in amenities, technological capabilities, comfort elements, and safety systems that enhance the driving and ownership experience.
Price (IDR)	The vehicle's monetary value expressed in Indonesian Rupiah, representing the seller's asking price and serving as a starting point for potential negotiations, reflecting market conditions and the car's overall worth.

TABLE II
EXAMPLE OF FINE-TUNING FORMAT

Role	Message
System	CarFin is a smart car recommendation assistant.
User	What do you think about the Suzuki bolan?
Assistant	Suzuki bolan is a 2013 car by Suzuki with 123456km mileage. It uses Petrol and has a Manual transmission. The car is priced at Rp 47,202,550 with features AM/FM Radio, Front Speakers, USB and Auxillary Cable.

D. Prompt Engineering

The prompt engineering process is carried out to design effective messages between the system and the user in the car recommender system based on Large Language Model (LLM). In Table III, the prompts are designed to guide the model in providing responses that are appropriate and relevant based on the input given by the user. With this technique, the system can provide more personalized and targeted recommendations. Proper prompt engineering enables the model to better understand the context and produce more accurate outputs in responding to user requests related to the car they are looking for.

E. Implementation

By utilizing LLM, in addition to ensuring that the interaction between users and Carfin becomes more interactive, the

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TABLE III
PROMPT ENGINEERING GUIDELINES FOR CAR RECOMMENDER

Role	Content
System	You are a bilingual (Indonesian and English) car recommendation assistant. Your primary function is ONLY to provide car recommendations and handle car-related queries. The system must always structure responses with specific criteria: Car Name: Toyota Camry 2.5G Year: 2020 KMs driven: 15,000 km Fuel: Petrol Transmission: Automatic Description: Well-maintained, accident-free Car Features: Sunroof, leather seats, navigation system Price (IDR): IDR 500,000,000 If the message is about car, provide specific car recommendations based on their needs include key features and keep responses concise but informative. But, if the message is not about cars, provide a predefined response "Sorry, I can only help you with car recommendations. Please ask about the car you are interested in."

applied fine-tuning and prompt engineering processes allow the system to provide specific and relevant car recommendations tailored to user needs.

We use Telegram as the main platform to facilitate interactions between users and Carfin. The app's welcome page, shown in Figure 2, features a professional design with a background of car rows and the Carfin logo. This page provides an explanation of the bot's main function as a car recommender system that assists users based on model, transmission type, and budget.

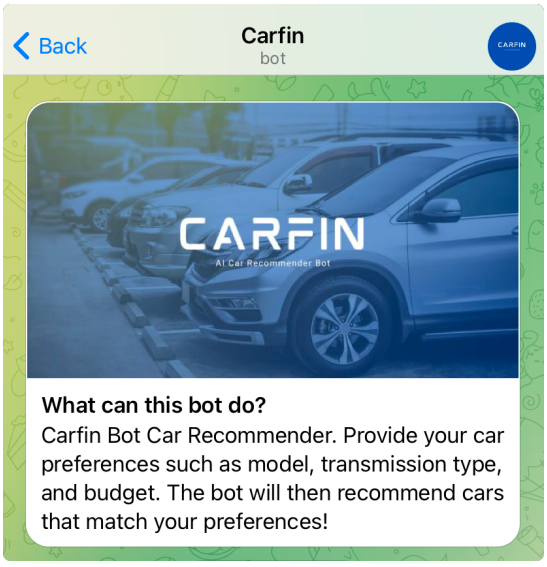


Fig. 2. Initial bot interaction

The initial conversation with Carfin begins by introducing itself and asking the user for preferences, such as their budget. Once the user provides the budget, the bot responds by recommending cars with detailed specifications such as the car's name, year, mileage, fuel type, transmission type, description, key features, and price, as shown in Figure 3.

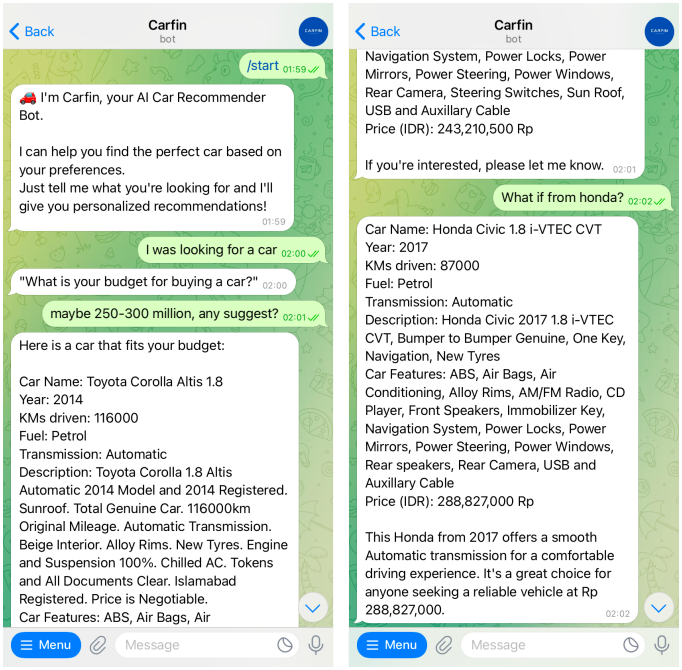


Fig. 3. Initial conversation

The bot's capability is also evident in handling specific requests. However, when users ask about topics outside the car domain, such as restaurants, the bot asserts its limitations with an appropriate response, as shown in Figure 4. This focus highlights Carfin's specialization in providing car-related recommendations.

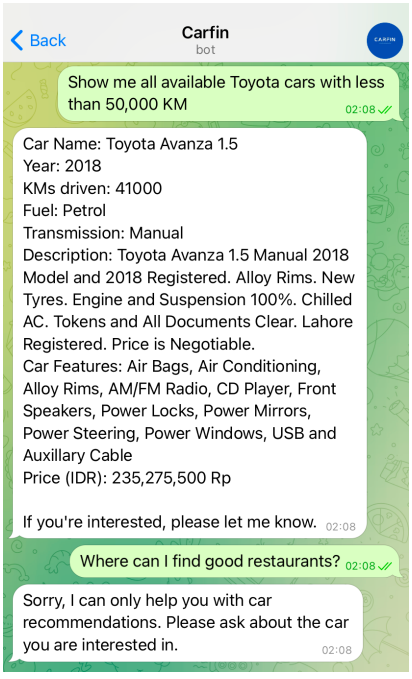


Fig. 4. System focus and boundaries

IV. RESULT AND DISCUSSION

This section discusses the results of fine-tuning and the evaluation of the car recommendation model. To measure the model's performance during the training process, the metrics

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training loss and validation loss were used. Training loss is used to evaluate the model's prediction errors on the training data, while validation loss is used to assess the model's generalization ability on data not involved in training. As shown in Figure 5, the graph indicates a significant decline in loss during the early phases of training, followed by stabilization as the training process progresses. The validation loss being lower than the training loss at the end of the process indicates that the model did not experience overfitting and is able to handle new data effectively. For evaluating the model's performance after fine-tuning, five key metrics were used BERT Precision, BERT Recall, BERT F1-Score, and Cosine Similarity.

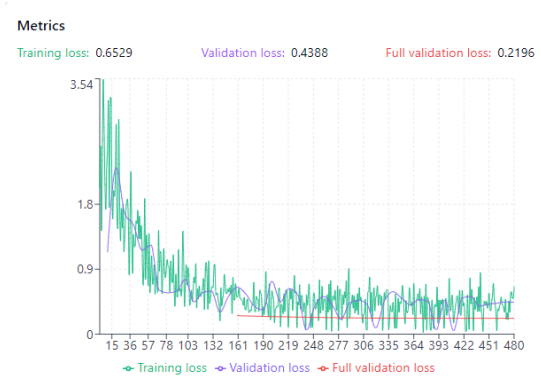


Fig. 5. Result of fine-tuning

BERTScore is used to measure the relevance of candidate text against reference text, calculating word embeddings using the BERT model to assess the quality of recommendations based on precision, recall, and their combination. The formula for calculating BERTScore is as follows:

$$BERT(\langle x_1, x_2, \ldots, x_n \rangle) = \langle x_1, x_2, \ldots, x_n \rangle \quad (1)$$

$$BERT(\langle \hat{x}_1, \hat{x}_2, \ldots, \hat{x}_m \rangle) = \langle \hat{x}_1, \hat{x}_2, \ldots, \hat{x}_m \rangle \quad (2)$$

To measure the similarity between two text embeddings, Cosine Similarity compares the angle between the vectors in a multidimensional space. The formula is explained as follows:

$$similarity(x_i, \hat{x}_j) = \frac{x_i \cdot \hat{x}_j}{\|x_i\| \|\hat{x}_j\|} \quad (3)$$

Next, to measure how well the candidate text covers information from the reference text, this is evaluated using the BERT-Precision metric. The formula is explained as follows:

$$P_{BERT} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \underbrace{cosine\ similarity(x_i, \hat{x}_j)}_{greedy\ matching} \quad (4)$$

Conversely, to evaluate how much information from the reference text is covered by the candidate text, the BERT-Recall metric is used. The formula is explained as follows:

$$R_{BERT} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \underbrace{cosine\ similarity}_{greedy\ matching}(x_i^T \hat{x}_j) \quad (5)$$

To provide a more comprehensive assessment, the BERT-F1 metric is used as the harmonic combination of BERT-Precision and BERT-Recall. The formula is explained as follows:

$$F_{BERT} = 2 \times \frac{P_{BERT} \times R_{BERT}}{P_{BERT} + R_{BERT}} \quad (6)$$

This evaluation provides a comprehensive understanding of the model's ability to deliver accurate and relevant responses, while also ensuring the model's generalization quality on new data. The evaluation results will be compared with the initial model's performance to demonstrate the impact of the fine-tuning process.

TABLE IV
COMPARISON OF MODEL PERFORMANCE BEFORE AND AFTER FINE-TUNING

Metrics	Pre Fine-tuning	Post Fine-tuning
Cosine Similarity	0.6590	0.8372
BERT Precision	0.8048	0.8541
BERT Recall	0.8639	0.9183
BERT F1-Score	0.8332	0.8850

The evaluation results show a significant performance improvement after fine-tuning the model. As shown in Table IV, the fine-tuned model achieved improvements across all evaluation metrics. This performance increase indicates that the fine-tuning process successfully optimized the model to better understand and provide recommendations within the car domain.

V. CONCLUSION

This study develops a car recommender system based on a Conversational Recommender System utilizing a Large Language Model. The evaluation results show a significant performance improvement after the fine-tuning process, as evidenced by the increase in the BERT Precision metric (from 0.8048 to 0.8541), BERT Recall (from 0.8639 to 0.9183), BERT F1-Score (from 0.8332 to 0.8850), and Cosine Similarity (from 0.6590 to 0.8372). The system successfully provides personalized car recommendations through natural interactions with users while maintaining its specialization focus within the car domain.

Future research for Carfin could focus on several key improvements. The system could be enhanced with real-time market data integration to provide up-to-date information on car prices and availability. The conversation model could be improved to better understand complex user preferences through longer dialogues. Adding image processing capabilities would allow users to analyze car photos and specifications. Regional market adaptation could help serve different geographical areas with specific car preferences. Integration with dealerships would enable direct connections to available inventory. Additionally, future research could focus on testing user experience by conducting direct trials with users to assess how easy the system is to use and how effective it is in providing recommendations. These

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enhancements are expected to strengthen Carfin's ability to deliver more practical and personalized car recommendations while meeting the evolving needs of the market.

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